When the video signal is digitized in the frame grabber, it is converted back into columns. However, these columns in the digitized image have *no relation* to the columns in the CCD. The number of columns in the digital image depends solely on how many times the frame grabber samples each line of video. For example, a CCD might have 800 wells per row, while the digitized image might only have 512 pixels (i.e., columns) per row.

The number of columns in the digitized image is also important for another reason. The standard television image has an **aspect ratio** of 4 to 3, i.e., it is slightly wider than it is high. Motion pictures have the wider aspect ratio of 25 to 9. CCDs used for scientific applications often have an aspect ratio of 1 to 1, i.e., a perfect square. In any event, the aspect ratio of a CCD is fixed by the placement of the electrodes, and cannot be altered. However, the aspect ratio of the digitized image depends on the number of samples per line. This becomes a problem when the image is displayed, either on a video monitor or in a hardcopy. If the aspect ratio isn't properly reproduced, the image looks squashed horizontally or vertically.

The 525 line video signal described here is called **NTSC** (National Television Systems Committee), a standard defined way back in 1954. This is the system used in the United States and Japan. In Europe there are two similar standards called **PAL** (Phase Alternation by Line) and **SECAM** (Sequential Chrominance And Memory). The basic concepts are the same, just the numbers are different. Both PAL and SECAM operate with 25 interlaced frames per second, with 625 lines per frame. Just as with NTSC, some of these lines occur during the vertical sync, resulting in about 576 lines that carry picture information. Other more subtle differences relate to how color and sound are added to the signal.

The most straightforward way of transmitting color television would be to have three separate analog signals, one for each of the three colors the human eye can detect: red, green and blue. Unfortunately, the historical development of television did not allow such a simple scheme. The color television signal was developed to allow existing black and white television sets to remain in use without modification. This was done by retaining the same signal for brightness information, but adding a separate signal for color information. In video jargon, the brightness is called the *luminance signal*, while the color is the *chrominance signal*. The chrominance signal is contained on a 3.58 MHz carrier wave added to the black and white video signal. Sound is added in this same way, on a 4.5 MHz carrier wave. The television receiver separates these three signals, processes them individually, and recombines them in the final display.

Other Image Acquisition and Display

Not all images are acquired an entire frame at a time. Another very common way is by **line scanning**. This involves using a detector containing a one-dimensional array of pixels, say, 2048 pixels long by 1 pixel wide. As an object is moved past the detector, the image is acquired line-by-line. Line